CHAPTER 1

# The Challenge of Climate Change for Future Port Cities: Considering the Precautionary Principle in Adaptation to Sea-Level Rise

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## 1 Introduction

Port cities have been transformed. While once they might have been narrated as sleepy, slow-paced, romantic places with sailing ships nestled on limitless waterscapes, they are now centres of commerce, trade, tourism, entertainment, and residential living. To borrow from Simmel (1903[1997]), port cities are the seat of the money economy where daily, they symbolise a "gate to the world and a place for the exchange of people, ideas, and goods" (Kowalewski, 2021, p. 55). In 2020 the global pandemic dramatically changed this situation. Once hustling and bustling port cities reliant on tourism were substantially impacted, as the flow of humans/non-humans reduced to a trickle, turning port cities into cathedrals of stillness. With the arrival of vaccines and the removal of state border closures, they are now experiencing economic/social/cultural recovery (Hein & Schubert, 2021a; 2021b). In the background, however, climate change continues to loom as a challenge to that port-city resilience.

Despite warnings for years about the potentially devastating consequences of climate change, the global community has been slow to mitigate the disaster. Annually, we are presented with more information about tipping points, as the world shifts increasingly towards an unavoidable two-degree warmer world. In these conditions, the continued unencumbered existence of ports is threatened by problems like sea-level rise, increased intensity of natural disasters (such as cyclones), and increased wave heights (Hurlimann et al., 2014). Port cities are threatened by these changes and require significant planning for sustainable futures. With the acceptance of unavoidable climate change and impacts like sea-level rise on the horizon, bodies such as the international port city association, AIVP, have written into their 2030 agenda a commitment to plan for and act now to make sustainable choices for port cities. This commitment is also known as climate change adaptation (Hurlimann et al., 2014).

Although all cities face problems related to climate change, port cities are particularly vulnerable (Nursey-Bray et al., 2013), despite their historical

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resilience to economic, social, and natural disasters (Schubert et al. 2022). As noted by several researchers, a significant proportion of the global population resides in coastal spaces, with an estimated 150 million projected to live below high tide lines by 2050,[1] and 360 million "threatened by annual flood events in 2100" (Kulp & Strauss, 2019, p. 2). Furthermore, Nicholls (2011) estimates that a significant portion of those vulnerable to sea-level rise already face significant inundation events annually.

As Shove (2010, p. 283) argues, the strategy for approaching the "specific challenges of responding and adapting to climate change" needs to shift towards how to "conceptualize the relative fluidity and fixity of what have come to be ordinary, but ultimately untenable ways of life." While her attention is directed towards issues of "changing patterns of mobility, food consumption, heating, cooling" and in general "green consumption" (2010, p. 283), in the case of port cities, the focus is solely on the viability of continuing the status quo. This can be "very difficult, given the high degree of incalculable dangers, or uncertainties" that exist in these spaces (Hein & Schubert, 2021a, p. 239). Furthermore, port cities are economically, culturally, and politically important to the nation-state, creating barriers to changes in soft and hard structures (Barnett et al., 2015; Hein & Schubert, 2021a, 2021b; Nursey-Bray et al., 2013). Even when decisions are made, the potential for maladaptation to future climate events is always a danger (Barnett & O'Neill, 2010).

This chapter seeks to address this question directly by questioning the mentalities that are embedded in the future orientations towards these places. In this chapter, I use the terms port cities/regions together as this is indicative of the relationship that port cities have with other spaces, both inland and up and down adjacent coasts. Port cities provide flows of income to regions, and as such, we need to consider the relations between the port city and the region surrounding it. We need to consider the 'hustle and bustle', or in other words the economic, cultural, and social acceleration of port-city life, in the face of impending sea-level rise. As I will argue, in this context a path dependency has been constructed through the port city, especially economically, that places specific limitations on planning for the future.

1 This is in a modelled scenario where there is a moderate response to climate change through mitigation. If there is a significant decline in carbon emissions, sea-level rise may well be much lower, or conversely, a business-as-usual model (high carbon emissions) over the next 30–50 years will result in potentially high sea-level rise – which would raise the number of those at risk. However, there are those who argue that the levels of increase in sea-levels is uncertain because of the difficulties of modelling with respect to ice loss (Rahmstorf, 2010; see this chapter).

In this chapter I first explore temporalities and futures, drawing on the work of Adam and Groves (2007) and Urry (2016). I will argue that our contemporary port cities/regions are the aggregated results of past generations' goals, ambitions, and importantly, economic desires. In understanding this, we need to unpack the question of what the current mentality towards port cities/regions is. Here I argue that capitalism especially accelerates the growth and importance of mobile spaces (such as ports) (Urry, 2016). The rationalised goals and ambitions of capitalists in the past have created a present that is experienced today in port cities. Adam and Groves (2007) describe this as the promethean power of socio-economic decisions that seek to colonise futures for gain. It is through this power, I argue, that the intensification of the economic importance of port cities restricts potential adaptation solutions into the future. A genuine attempt to address adaptation requires consideration of what we ought to do. With the help of Shaw (2010), I argue that an ethics of responsibility through the precautionary principle needs to be considered in our present-day relationship to port cities/regions. Specifically, we need to reimagine the temporality of port cities from one of increasing tempo to one of potential slowness and sustainability, adapting to the conditions of climate change while potentially reinvigorating the place-value of port cities.

## 2 Temporalities, Futures, and Promethean Power

The transition from premodernity to modernity and the changes to how we have understood the future are important to foreground. The sea, relevant to our discussions, once held a divine, sacred quality which could both create and destroy, as witnessed in the deluge mythologies of Judeo-Christianity (Osbaldiston, 2018). Oceans were controlled by a God, who held them in check after the post-diluvian promise. The natural rhythms, flows, and importantly, disasters conjured by the seas, were not to be understood by humans. Even to look upon the deep and dark unknown of the ocean encouraged a sense of unease and anxiety amongst premodern believers.[2]

Pandora's box was opened through the Enlightenment scientific revolution. The management, measurement and calculative nature of rational science

2 As several scholars note, seas in the Old and New Testaments are often places that are metaphors for the unknown. For instance, the famous passage of Job 38:8 declares "or who shut up the sea with doors, when it brake forth, as if it had been issue out of the womb?" As several commentators have noted, the sea was the domain of God (and devils), managed, as it were, by divine control out of reach of humanity (Osbaldiston, 2018, p. 44–46).

did away with a supernatural explanation of the natural world. The transition of the modern institution from irrationality towards modern instrumental rationality (as stipulated famously by Weber 1919[2012]) set the scene for a transformation of how we understand the future – and the sea. Using Weber's framework, Adam (2009, p. 9) writes the following,

> [The] rise of scientific knowledge and the socio-economic capacity to apply a rational calculus to ever-expanding spheres of social life, people were able to transcend the socio-economic present and their will could be imposed on both the personal and collective future. The future, therefore, was no longer a mere continuation of the past but became ever more a consequence of action in the present.

The world for Weber (1919[2012]) was now observable and rationalised. Importantly, the future was calculable and no longer simply unknown or in the hands of fate or the Gods. Rising out of this transition were instruments and knowledges capable of predicting, holding stable, and utilising time for the purposes of rational goals. Disciplinary knowledge such as economics, cartography, engineering, geography and even sociology, emerged to create opportunities for the capturing of futures for capital gain. As Adam and Groves (2007) contend, capitalism's core promise is, after all, investing now for future returns.

Rationalisation constructed new pathways and opportunities, but also unintended consequences. It allowed for a future that could be calculated, predicted, and traded through the measurement of time, money, and risk (Adam & Groves, 2007). Decisions made in the present based on the predictive power of probability could be made for the direct purpose of capturing some future state. For example, investment in a port in one area, which would reduce time for transportation between a mine inland and transport ships, could lead to profit in the future. Time calculated in this manner became money (Adam, 2009).

From this perspective then, the future becomes "emptied of content" allowing one to rationalise, make knowable, and plan for times when the present will pay off (Adam & Groves, 2007, p. 13). Trade began in earnest through merchants, as evidenced by the growth of the East India Company, for example. "Global merchants in complex calculations about potential profit and loss over long periods, with goods being traded in absentia" became a hallmark of Western European capitalist enterprise in early modernity (Adam & Groves, 2007, p. 13). As statistical analysis and probability theory expanded, the capacity to make money by trading on futures increased and the rise of rational

instruments like insurance provided safety nets for calculable losses (Adam & Groves, 2007).

This transition to a world where futures are traded through economic activity (or risks) relied heavily on rational science to hold things stable or modelled so that effective calculations can take place. Of course, as several scholars have argued in recent decades, the growth of this managed future created unknowable side effects or unintended consequences (Adam & Groves, 2007; Adam, 1995; Beck, 2009; Urry, 2016). Capitalism and the securing of futures economically has in part created a wicked problem for us today through climate change (Urry, 2016). However, as Chakrabarty (2009, p. 221) argues, this is not simply capitalism's fault. Our efforts to control the natural world, ensure future security (such as through energy), and improve the conditions of life generally have led to the development of large-scale problems (Urry, 2016). Urry (2016, p. 188) contends that our pasts are replete with complex and interdependent sociomaterial systems, that "stretch over time-space" and formulate our present today. In our attempts to fix different problems, we have created new ones, climate change being one of the most significant today (Urry, 2016, p. 188).

The mentality that conceives of the future as something empty and ready to be conquered for the purposes of economic or social gain is described by Adam and Groves (2007) as promethean power.[3] In short, this is "evident in contemporary knowledge practices in science, technology, politics and economics" which "construct and produce futures without being able to know and take responsibility for the consequences" (Adam & Groves, 2007, p. 201). In other words, one's choices in the past that transform futures are devoid of responsibility because one cannot be held responsible for knowledge of things yet to pass. The "institutional sphere of responsibility' is often shifted away from the originators of the timeprint (who are in the past), and onto mostly "successor generations" or in the case of sea-level rise, onto under-developed nations (Adam & Groves, 2007, p. 203).

It is important to recognize that this time-space imprint is part of our everyday lives. To understand our present conditions, we need to interpret past "social ideals, rationalities, values, morals, and desires" and how this reaches through time and space imprinting itself onto our present (Osbaldiston, 2018, p. 227–228). Our institutions, infrastructures, places, spaces, and relations to

3 Prometheus was a lesser God who stole fire to provide to the people of Earth – despite the other Gods feeling it to be improper as they considered "humans not prudent and restrained enough to handles this precious gift with wisdom" (Adam & Groves, 2007, p. 232). Other accounts of the tale suggest otherwise; however for the purpose of this chapter, Adam and Groves' approach is used.

our natural worlds are the result of generations of decisions, constructing powerful systems and networks, which have cascaded down through history into our present day (Urry, 2016, p. 72).

These present-day conditions were not considered by our forebears, nor were they predictable. The framing of the future as something that can be calculated and known relies on knowledge of things as they are (held constant) and not of things that are unknown. Promethean power means taking away obligation to the future. Adam and Groves (2007, p. 81) write,

> [T]his perspective concentrates attention on the present at the expense of history and futurity, with the effect that we are no longer able to grasp our products with reference to their social origin and socio-environmental destiny. Power becomes promethean in the sense that the power to act and transform is not matched by a capacity to know and be mindful of interconnections, implications, and potential effects.

This creates an inability to be able to predict potential side effects, and therefore an inability to think about the implications of our choices for future generations (Urry, 2016). Doing so involves thinking through problems and arriving at solutions meant especially for our forebears, fixing data and knowledge, and then predicting outcomes accordingly. However, as others like Beck (1992) show us, these processes can never account for unknown systems or unpredictable data.

In contemporary times we still trade on futures. This is perhaps no more evident than in the modern port city where the hustle and bustle of economic life is underpinned by futures via the movement of goods, services, and people. Yet, climate change challenges this hustle and bustle, requiring a rethinking of choices in the present through an obligation to those of unborn generations. Adam and Groves (2007, p. 237) write, "as owners of the future we also carry the sole responsibility for the outcomes of our future-creating actions." From this perspective, our actions, which will shape the choices of future peoples, make us "inescapably responsibly for that which we cannot know" (Adam and Groves, 2007, p. 237). Unfortunately, the ability to construct an ethics of responsibility for things that we do not know is almost impossible institutionally. Port cities are locked in an economic system that depends on its certainty. We literally trade in these places on future economic payoff, by holding the port city as predictable and reliable. However, climate change and in particular sea-level rise and the associated risks related to it create a future that is relatively unpredictable; planning to maintain the status quo is difficult (Christodoulou, Panayotis and Demirel, 2019).

## 3 Port Cities, Promethean Power, and Climate Change Adaptation

The preceding discussion may seem slightly odd when examining port cities. However, they are examples of promethean power par excellence. Ports themselves, represent a desire to trade on futures. Nation-states in 17th century Europe looked to the seas through new forms of economic activity that relied on futures traded. Adam and Groves (2007, p. 10) write,

> The trade in futures buys and sells futures for the benefit of the present, that is, for profit in here and now. It trades not just in goods but the promise of goods. This was of particular importance to international trade by sea given that trade ships might be away for as long as three years at a time. It involved merchants in complex calculations about potential profit and loss over long periods, with goods being traded in absentia.

Ports were and are vital to the ability to trade these futures to capture profit in the present. They are now hubs of rational economic activity, calculable and instrumental to the nation-state.

Ports represent a past generations' desire for wealth – and the trading of futures via shipping. Over time, what Weber (1915[1970]) described as an instrumental rationality became embedded in ports not just organizationally, but also in the materiality of the port and surrounding natural environments. This instrumental rationality, or trading based on maintaining a calculative certainty over the future, is what we could describe as the *path dependency* of port cities/regions. Described largely in terms of government, policy, and bureaucratic decision-making, path dependency "characterizes specifically those historical sequences in which contingent events set into motion institutional patterns or event chains that have deterministic properties" (Mahoney, 2000, p. 507). Once these patterns or events are set into motion, they are difficult to break down or reverse due to their "self-reinforcing properties" (Mahoney, 2000, p. 512).

In the case of ports, among the most significant of these self-reinforcing properties are the very economic and social frameworks that sustain them perpetually. To use Adam and Groves' (2007, p. 11) words, ports are sustained by "socio-economic activities that seek to shape, form and colonise a future of our own making." We are locked into an economic system especially that wholly depends on the maintenance of ports. For instance, in Australia, an island continent rich with natural and agricultural resources, roughly 98 per cent of trade is shipped through the sixty international trading ports across the country (Ports Australia, 2022). This sustains a significant portion of the economic

activity not just within ports and their associated regions/cities but also in other areas such as transport/logistics, which supplies around 15 per cent of Australia's gross domestic product (GDP) (Nursey-Bray et al., 2013, p. 1023).

As noted by Hein (2021, p. 3), port cities are no longer bounded spaces but are increasingly porous as a result of the construction of infrastructure, development, commercial spaces, and even residential areas in and around the port itself. Ports, though economically important to city authorities, were not prioritised by planners who sought to address "local economic, social, and other issues" (Hein, 2021, p. 5). As Hein (2019, p. 4) argues elsewhere, as a result, port cities are now "also large consumer hubs where actors pursue many non-maritime or non-port orientated economic, ecological, social-cultural and spatial goals." Furthermore, port cities "emerge as strong economic, political, and cultural centres that facilitate trade and travel" (Hein, 2019, p. 4).

The result of this activity and the expansion of trade is the hustling and bustling port cities we see today. The port city environment is one of non-stop activity. As Urry (2016) suggests, these places have emerged through the collision of different socio-material networks (e.g., global trade, construction, education, residential accommodation), coalescing to produce mobile, constantly growing economic and cultural hubs (see also Hein, 2019). Tourism is a prime example of one of these networks that adds to the intensification of fast-paced life and the movement of people in and around port cities and regions. Tourism relies heavily on access to the water, often through the port itself, or through surrounding areas. Cruise ship terminals and the traffic that travels through them, not just people, enable an intensification of economic, social, and cultural exchange (Hein, 2021; Kowalewski, 2021). Economically, these are vital contributors to a local region. For instance, in 2004 it was estimated that in the smaller regional city of Cairns in Far North Queensland, Australia, a port call of a cruise liner would add over $240,000 (AUD) per day to the local economy (Dwyer, Douglas & Livaic, 2004). Pre-COVID-19, the cruise ship industry was continually touted as the fastest-growing tourist industry, globally generating almost $18 billion annually (Brida & Zapata-Aguirre, 2009).

Such an injection of capital outside the usual port activities of delivering cargo only serves to intensify the hustle and bustle of port-city life. However, it also means that many port cities/regions are highly dependent on these industries. Cairns for instance, which ships approximately 1.3 million tons[4] of cargo a year through its port (Ports North, 2022), is invested heavily in domestic and international tourism (including cruise shipping). Prior to the pandemic,

4 This was the financial year output for 2019. 2020/2021 data is yet to be made available at the time of this publication (Ports North, 2022).

the regional port city attracted approximately over 10 million visitor nights in 2018/2019, roughly half from international tourism (Tourism Research Australia, 2021). In 2019 the international tourist expenditure in the Far North Queensland region was estimated at $3.9 billion (AUD), representing one of the leading contributions to local economic conditions (Tourism Research Australia, 2021). COVID-19 shut tourism there down, causing significant angst amongst the community and local leaders. A once vibrant cityscape transformed almost overnight into a still and anxious space.

Of course, Cairns is not the only place in Australia and the world to be impacted by the sudden shutdown of international tourism by the pandemic. In Australia alone, this event led to a loss estimated around $42 billion in GDP along with direct job losses of over 150,000 in tourism industries and a further 450,000 in the tourism chain (Pham et al., 2021). However, what COVID-19 and the prevailing shutdown of tourism exposed were the path dependencies of local economies, especially in port cities/regions. We became more thoroughly aware of the vulnerabilities of these spaces even though the impact was relatively short, and for the most part tourism is starting to return a few years later.

What the pandemic also revealed is how port cities are planned, designed, and constructed with futures in mind. Infrastructure in these spaces is constructed, in relation to tourism, for the purposes of entertainment, consumption, accommodation, and access to other tourist sites connected to port cities. However, the port city itself is the result of the past. Ports, especially in colonial states such as Australia, were places for transportation of goods and people, but cities emerged over time to sustain them while also developing their own intense economic, social, and culture life (Hein, 2019; Palmer, 2020). The port city today again represents the meeting of disparate but connected networks of economic and social exchange, to present what is now a wicked problem for nation-states to deal with.

This wicked problem includes the significant infrastructure located in coastal areas. Port infrastructures such as warehousing, storage facilities, industrial zones, and non-port infrastructures such as urban housing, tourist amenities, commercial zones, and so on are all in danger of future inundation through coastal hazards (Christodoulou, Christidis, & Demirel, 2019; Hein & Schubert, 2021a, 2021b; Osbaldiston, 2018). The development and subsequent precarious position of this infrastructure are what we might deem the unintended consequences of the past. While planning can account for foreseeable dangers in the present (such as cyclone risks), it is difficult to assess future dangers with the unpredictable nature of sea-level rise. Maintaining the status quo of the port city is increasingly challenging with current planning regimes.

Sea-level rise and "changes to precipitation [...] sea surface temperatures, ocean currents, ocean acidity, extreme weather events and tropical storms" and changes to wave heights and inundation will have potentially devastating impacts that cannot be held constant and planned for (Hein & Schubert, 2021a, 2021b; Nursey-Bray et al., 2013, p. 1029). The vulnerability of port cities is the culmination of a host of unintended consequences of future taking which include the promethean power of decisions to use greenhouse gases in the past to achieve economic progress, along with the ambitions of global capitalism that led to the growth of massive cities perched on the shoreline (Osbaldiston, 2018).

While the impacts of climate change could not necessarily be anticipated by past generations, nor the idea of sea-level rise even be considered, numerous events have occurred historically when the seas have shut down cities via natural disasters (Hein & Schubert, 2021a; Osbaldiston, 2018). Despite these events, the fervent need for economic and social progress has been sustained throughout modernity – a mentality that persists in *rebuilding, protecting, and progressing*. Port cities have a type of coastal amnesia towards past events, choosing instead to think that we can continue living, developing, and investing our economic and social futures in coastal spaces. In short, and to continue the theme of this book, we seek to continue intense development, encounters, events, flows, and above all, economic activity in these port cities, which have latent impact on regions that surround them.

The major difficulty with sustaining this mentality now, as we know, is that climates are and will continue to dramatically challenge our assumptions about coastal living (Osbaldiston, 2018). The institutions tasked with the governance of our port cities/regions hold, I argue, a promethean view which seeks to progress socio-economically with predictive power that is limited at best. Political power is "embedded in the short-term calculative and thereby knowable futures that lead to examining choices based on cost-benefit analysis, legal precedents/ramifications and the predictive capacities of science (such as coastal flood modelling)" (Osbaldiston, 2018, p. 238). Unfortunately, preparing for future sea-level rise requires a longer-term thinking that escapes this approach. Institutions have developed a locked-in rationality towards the coast, limiting the capacity to think through potential unknowns. Urry (2016) contends that planning for the future by *knowing* the future is impossible. However, he also argues that scenario building in planning, using different scales of measuring outcomes, is a potential way forward (Urry, 2016, p. 97).

I argue that the inability to hold certain or know the future is a functional and intellectual barrier to climate change adaptation in port cities worldwide. Barnett et al. (2015, p. 5) argue that "history shapes current decisions, and the

changes identified for adaptation are impeded because they work against existing governance institutions." Valente and Veloso-Gomes (2020, p. 405) further contend that port cities will need to be proactive in their approach to coastal risks, planning for "longer temporal horizons, consider[ing] the dynamic and uncertain nature of coastal systems and risks." However, the options for dealing with sea-level rise and associated risks are already limited because of past decisions to position port cities so close to the shore. Adaptation to sea-level rise in planning is commonly now framed through four choices: managed retreat of infrastructure and communities inland, protection against sea-level rise which uses hard and soft barriers (e.g., sea walls, coastal vegetation), a mixture of both options, and a do-nothing approach resulting in future unplanned adaptation (Cooper & Lemckert, 2012). These options have costs and limitations. The first three all rely on one very significant assumption, that sea-level rise can be calculated, modelled, and planned for. The final approach, however, risks loss of significant property making the port city potentially unviable. Whatever the choice, our planning decisions shape potential choices in the future.

## 4 Port Cities, Futures, and the Precautionary Principle

The difficulty of climate change adaptation in port cities lies in the changes required to the way they are arranged, organized, and future-orientated (Barnett et al., 2015). Positioning and planning these potential adaptation strategies means invoking some policies that are proactive and not reactive, but also open to different scenarios, as climate science on sea-level rise is rather uncertain (Osbaldiston, 2018). Adam and Groves (2007, p. 185) sum up the situation in the following,

> The difficulty confronting us [...] is that the indeterminacy of unbounded effects makes reliance on scientific prediction and economic risk calculation inappropriate and presents us instead with questions about justice, rights, and possible harm to future beings that have to be addressed.

The difficulty confronting the challenges of sea-level rise is that the unbounded effects can only be planned in current regimes if we can adequately predict how high the seas will rise (Valente and Veloso-Gomes, 2020). The inadequacy of current planning systems in port cities is two-fold, however. Firstly, the capacity to begin adapting now for something into the future will still bear an economic cost (such as developing larger seawalls or protecting coastline not yet protected) and may well still have unintended consequences if not

carefully engineered (Barnett & O'Neill, 2010; Osbaldiston, 2018). Secondly, even this option requires some degree of certainty over the actual height of the seas in the future. Planning needs to accept the uncertainty and become 'adaptive', while also acknowledging that the status quo of port-city life may not be sustained (Valente & Veloso-Gomes, 2020).

In relation to this latter issue, we see often in adaptation planning the holding of sea-level rise to a certain height. One of the barriers to adaptation is in fact the uncertainty involved (Butler, 2008). If we were certain and could calculate with precision where the seas would be, how often storms may hit, how far inundation would go inland, and the impacts on critical infrastructure, then adaptation may well progress urgently (Butler, 2008; Osbaldiston, 2018). We would have an exactness regarding sea-level rise that could be held stable and planned for.

In a recent review of sea-level rise modelling, Horton et al. (2020) highlight how uncertain calculating this future event is (see also Rahmstorf, 2010). Reviewing research calculating sea-level rise in the years 2100 and 2300, they highlight the fact that in 2100 alone, the modelling ranges from 0.3M to 1.65M (Horton et al., 2020, p. 18). The range is based on several scenarios, such as changes to our greenhouse gas emission rates. IPCC (Intergovernmental Panel on Climate Change assessments and predictions rely on this scenario building, ranging from zero-carbon futures to business-as-usual or the status-quo. In the case of the latter, sea-level rise is difficult to model given the uncertainties and "imperfect understanding" presented by the Antarctic and Greenland Ice Shelves which are "the largest potential contributors to GMSL (Global Mean Sea-levels)" (Horton et al., 2020, p. 5). Furthermore, Rahmstorf (2010, p. 45) has argued that in the past ice age, the "Earth slowly warmed by 4C to 7C" creating approximately a meter per year of sea-level rise.

This uncertainty makes planning for sea-level rise increasingly difficult for port cities. Choices will be required but planning without exactness is problematic. For instance, in Australia, some government planning agencies use as their constant the figure of 0.8M Australia Height Datum (AHD) in their coastal sea-level rise modelling (Hurlimann et al., 2014; Barnett et al., 2015). In port cities like Cairns, adaptation policies utilise this calculation to plan for vulnerable parts of the coast (Cairns Regional Council, 2021). Once the modelling is complete, short to long-term strategies are pursued to adapt to this level with higher sea walls, bunds, roads, flood pumps, and other infrastructure, along with soft adaptation strategies such as maintaining beaches through nourishment programs and increasing vegetation in certain vulnerable spaces (Cairns Regional Council, 2021). Challenging this approach, of course, is the potential for seas to rise even higher than 0.8M to 1.65M or

above. As Valente and Veloso-Gomes (2020) contend, this is not flexible or adaptive planning.

The decision by planners in this case to protect/defend rather than consider the managed retreat of their port city is yet another example of the promethean power of future making and a mentality that seeks to sustain the life of the port city as a place of hustle and bustle. This is not just about economic gain, though a major factor. Rather, the port city has become a vibrant space, full of culture, life, mobility, and social/cultural capital. As such, planning in this regard seeks to maintain this livelihood. However, I argue (see Osbaldiston, 2018) that doing so negates the potential to turn the port city into one of space, stillness, and openness. The port city does not need to remain ever increasingly on the move. Furthermore, holding sea-level rise constant eliminates potential scenarios and a range of possibilities not just in terms of vulnerability and danger, but also in terms of place-value and the intrinsic importance of coastal spaces to society.

Difficulties arise however when attempting to use future scenarios in adaptation, as certain events may not occur. As Cooper and Lemckert (2012, p. 12, *italics added*) argue, "predictions of *high sea-level* rise are subject to much uncertainty" which would then "keep such scenarios outside serious political debate and thereby preclude the development of adaptation plans" that are suitable. Important here is desire to "maintain certain essential elements without which the city loses its *raison d'être*" (Cooper & Lemckert, 2012, p. 2; Barnett et al., 2015). In the case of the port-city, risks to the heart of the place as one of a fast-paced economic and social hub through adaptation plans for future events, are unacceptable. Attempts to deal with a high sea-level rise (such as beyond 1.5M) would likely not receive political, financial, or public support (Valente & Veloso-Gomes, 2020). For instance, raising a seawall that could deal with high sea-level rise is likely to be unpalatable for certain port cities that have derived public value from seascapes and coastal amenities (Osbaldiston, 2018). Planning for significant high-risk scenarios also risks the port city losing its cultural value. So how do we deal with these wicked problems?

In the past, I have argued in relation to coasts that we need to open the mentalities/rationalities built into our systems today and redefine what orientation we have towards coasts into the future (Osbaldiston, 2018). Building on Wynne (2010) in particular, I argued for a "ecological or green" approach to coastal spaces via the collective will of the public (Osbaldiston, 2018, p. 249). Coasts could be subjected to a wave of green policies that would prioritise the value of the natural environment, provoking radical adaptation plans (such as managed retreat). Transforming land into common areas through a mixture of

retreat-and-buy back schemes may provide options for future planners to keep "social and economic" futures of certain locations in place for the moment, and through triggers or timelines, slowly remove vulnerable places in stages (O'Riordan et al., 2008, p. 154). In other words, land could be reclaimed as common property for the state, through adaptive and staged retreat planning.

The problem for large port cities is that these two options are highly unlikely to be adopted. Institutional change in adaptation, as noted by Barnett et al. (2015), is slow, perhaps too slow for the need to deal with sea-level rise now. More significantly, port cities are vital to national, local, and individual economic needs. They are hustling and bustling places for a reason – sustained by systems outside and within that require their continued high intensity operation. Dealing with this wicked problem also needs to recognize the social justice for those in the present who rely on port cities for their welfare. In what follows therefore I offer one potential solution that will require us to consider how we can shift or decentralise that hustle and bustle that is now found in port cities into the future.

One ethical approach we should adopt here is the precautionary principle (Shaw, 2010). Defined in the 1998 Wingspread Statement,[5] this ethical guideline articulates that "when an activity raises the threats of harm to human health or the environment, precautionary measures should be taken even if some cause-and-effect relationships are not fully established scientifically" (cited in Shaw, 2010, p. 104). Four key practical elements accompany this: taking preventative action despite uncertainty, placing the burden of harmful activities onto those who are creating it, exploring potential alternatives to possible harmful actions, and allowing further democratic decision-making in scientific advancement through the public (Shaw, 2010, p. 104). Shaw (2010, p. 104) contends in relation to climate change mitigation that we have gone past the ability to use "precaution" as we have entered unavoidable dangerous levels. The precautionary principle was not adopted appropriately as most nation-states adopted a "cost-benefit analysis" with short-term mentalities (Shaw, 2010, p. 118–119). The "precautionary window of opportunity has now passed" (Shaw, 2010, p. 118).

Planning for a port city that is challenged by climate change needs to embrace a new perspective. Shaw (2010, p. 119) suggests,

5 The Wingspread Conference was held in 1998 and combined the resources of scientists, legal experts, government officials, and environmentalists. It developed a statement that relied on the principle of "no harm" which has since been adopted by various institutions such as the European Union (Shaw, 2010).

> Rather than simply being content to avoid harm, we might instead begin developing positive goal-based visions of the future. The precautionary principle is thus reconstructed as something other than a process which is triggered by a crisis. Instead, the precautionary principle becomes an omnipresent, value-based, decision-making screen, which continually guides our decision-making process away from potential harms.

The point here is to balance short-term cost-benefit thinking with longer-term effects. A new narrative that envisages the future port city in an alternative way, in place-value and economic viability, could enhance current and future societies. This entails planning not through consultation with calculative experts alone but also with the public about what is valuable to the city and how we can avoid the future risks (Shaw, 2010; see also Wynne, 2010, p. 299).

Therefore, in relation to adaptation to the port city, the precautionary principle works in different capacities. Firstly, we must embrace the idea that holding sea-levels constant to plan for futures is fraught with danger. If we plan for small levels of sea-level rise, we risk not doing enough to hold the seas at bay, and thus we reduce choices for planners in the future. Any attempt to engage with adaptation here in the present needs to embrace the no-harm principle regarding future generations. Secondly, and conversely, the new narrative of the port city should embrace the precautionary principle in a positive way. Plans for port cities could be developed that engage citizenry and design new 'scapes' to maximise potential benefit, economically, and culturally. If we do want, for instance, to maintain a port city that is lively, attractive, bustling, and rich with 'life', then we need to potentially reimagine what that *scape* might look like.

What this requires, as Hein (2021) argues, is recognizing the port city as porous not only to the port itself, but also inland to the region. To adapt to sea-level rise, embracing a vision of the port-city as a region, linked socio-economically across spaces, is vital. This means adopting a new planning strategy that is adaptive, using "tipping points" for a future port city that gradually retreats inland into the region surrounding it (Valente & Veloso-Gomes, 2020). This includes the relocation of essential services (hospitals, government offices, emergency services, courts, etc.) to higher ground, putting in place buy-back schemes with incentives for commercial enterprise to shift into new inland estates, and turning coastal space into public/common land for leisure. These changes would transform the port city into something different. Amenity shifts from one of hustle and bustle, to a city that encourages slower activities and engagement with the natural aesthetic of coasts (Osbaldiston, 2018). The port city would still require tourist and commercial infrastructure, which in turn would require certain adaptive features to deal with potential

inundation. However, a decreasing intensity of movement inside port cities would mean less pressure on built defences of significant infrastructure (especially those essential services). Transforming the spatial environments could provide access for less dense tourist amenities – such as coastal parks – thus enhancing the place-value of the coastline.

Such an approach is unlikely in large port cities that have intense infrastructure that would be near-impossible to shift. As such, defences in these cases will need to be engineered, or allow for the flow of water in such a way that adds to port-city life. Potentially, these protections could enhance the city and restyle the direction and intensity of activity. Hein (2021, p. 5) describes for instance the "higher dykes" of Elbbrücken which "have been transformed on both sides into stepped terraces that allow strolling, outdoor activities, and sitting on the edge of the city overlooking the port." We could imagine similar defences in tourist hotspots like Cairns, where the built environment could enhance engagement with natural surroundings to create new attractions, new lifestyle opportunities for residents, and socio-economic advantages. With this approach, there is a danger of maintaining a status quo and creating serious unintended consequences. However, this might be the easiest option to sustain a port city/region that is highly dependent on a hustling/bustling mentality of economic activity.

Other approaches need to be considered as well. We need though to break open the promethean mentality and embrace the precautionary principle. It requires the citizenry to consider the uncertainty of sea-level rise and other associated issues and collectively seek a future that rises above self-interest (Wynne, 2010). Such a transformation of the public's everyday mentality seems difficult. However, we have seen across the world a rise in consciousness of the collective on the matter of climate change. A morality amongst citizens towards unborn generations has arisen and is perhaps the new mentality that must be tapped into so that we can adapt appropriately in port cities.

## 5 Conclusion

To uncertain invisible un/real and im/material
Deeps rippled through space-time-matter
Caring for strangers & fellow beings
Not yet existing but implicated
Our processes in progress
Changes in Practice & Responsibility

ADAM, 2018, p. 388

As the above meditation from Adam (2018) suggests, we are now writing the futures of generations to come, who will reside, work, govern, and importantly, be responsible for port cities/regions. In particular, the decisions we make regarding climate change adaptation in the present are rippling through time, creating socio-material realities. To embrace the future, we may need to alter our perception of the port city as a place of intense economic, social, and cultural activity. In short, perhaps our discussions need to be centred on turning port cities from places of hustle and bustle into places of relative slowness. Such a change alters our mentality of what port cities are for – adopting a new coastal ethic that envisages a port city as a calm and still place.

As argued, port cities themselves are locked in path dependencies: practices of institutions are geared towards an empty future to be used for economic and social progress. Port cities are now vital for the economic well-being of nation-states, local areas, and individual lives. Breaking this mentality apart in time to meet rising sea-levels is perhaps near impossible. An adaptation strategy needs to understand that people will still need justice in the present, as well as in the future.

What we will need to do however, as I have argued above, is embrace the precautionary principle in our adaptation plans in a manner that addresses the question of no-harm in a productive fashion. Transforming the hustle and bustle in a sustainable way, one that encourages the mixed use of retreat and defence, while also adding amenities to the port city in various ways, will be difficult. However, decluttering the port city and producing new coastal spaces via defences may well enhance place-value. Should we accept that the port city must sustain some of its *raison d'être*, we must also accept that relying on measurements or calculations of precision will potentially be misguided. As such, to embrace precaution means embracing uncertainty, and thus future-mindedness applies.

As I noted in the introduction of this chapter, the port city is now presently an interconnected hub of industrial, commercial, retail, residential, and tourist interest. The port city is a focal point of the money economy. However, does it need to be? Could we envisage a slower, less intensely developed and drastically redefined port-city space? Or conversely, could we envisage a port-city space that maintains this approach, with radically redefined spaces and activities in certain vulnerable locations? Either way, what we must appreciate is that our mentalities and choices about how we adapt are going to matter, immaterially and materially, to future generations. We cannot choose simply to ignore the threat of climate change to our port cities.

## References

Adam, B. (1995). *Timewatch: Social analysis of time*. Wiley.

Adam, B. (2009). Cultural future matters: An exploration in the spirit of Max Weber's methodological writings. *Time & Society, 18*(1), 7–25.

Adam, B. (2018). Four Meditations on time and future relations. *Time & Society, 27*(3), 384–388.

Adam, B., & Groves, C. (2007). *Future matters: Action, knowledge, ethics*. Brill.

Barnett, J., & O'Neill, S. (2010). Maladaptation. *Global Environmental Change, 20*(2), 211–213.

Barnett, J., Evans, L. S., Gross, C., Kiem, A. S., Kingsford, R. T., Palutikof, J. P., Pickering, C. M., & Smithers, S.G. (2015). From barriers to limits to climate change adaptation: Path dependency and the speed of change. *Ecology and Society, 20*(3), 5.

Beck, U. (2009). *World at Risk*. Polity Press.

Brida, J. G., Zapata-Aguirre, S., & Zapata-Aguirre, S. (2009). Cruise tourism: Economic, socio-cultural and environmental Impacts. *International Journal of Leisure and Tourism Marketing, 1*(3), 205–226.

Butler, C. (2008). Risk and the future: Floods in a changing climate. *Twenty-First Century Society, 3*(2), 159–171.

Cairns Regional Council. (2021). *Our Cairns Coast*. Retrieved from https://www.cairns.qld.gov.au/community-environment/climate-change/our-cairns-coast.

Chakrabarty, D. (2009). The climate of history: Four theses. *Critical Inquiry, 35*(2), 197–222.

Christodoulou, A., Christidis, P., & Demirel, H. (2019). Sea-level rise in ports: a wider focus on impacts. *Maritime Economics & Logistics, 21*, 482–496.

Cooper, J. A. G., & Lemckert, C. (2012). Extreme sea-level rise and adaptation options for coastal resort cities: A qualitative assessment from the Gold Coast, Australia. *Ocean & Coastal Management, 64*, 1–14.

Dwyer, L., Douglas, N., & Livaic, Z. (2004). Estimating the economic contribution of a cruise ship visit. *Tourism in Marine Environments, 1*(1), 5–16.

Hein, C. (2019). The port cityscape: Spatial and institutional approaches to port city relationships. *PORTUSplus, 8*(Special Issue), 1–8.

Hein, C. (2021). Port city porosity: Boundaries, flows and territories. *Urban Planning, 6*(3), 1–9.

Hein, C., & Schubert, D. (2021a). Resilience, disaster and rebuilding in modern port cities. *Journal of Urban History, 47*(2), 235–249.

Hein, C., & Schubert, D. (2021b). Resilience and path dependence: A comparative study of the port cities of London, Hamburg, and Philadelphia. *Journal of Urban History, 47*(2), 389–419.

Horton, B. P., Khan, N. S., Cahill, N., Lee, J. S. H., Shaw, T. A., Garner, A. J., Kemp, A.C., Engelhart, S.E., Rahmstorf, S. (2020). estimating global mean sea-level rise and its uncertainties by 2100 and 2300 from an expert survey. *npj Climate and Atmospheric Science, 3*(1), 18.

Hurlimann, A., Barnett, J., Fincher, R., Osbaldiston, N., Mortreux, C., & Graham, S. (2014). Urban planning and sustainable adaptation to sea-level rise. *Landscape and Urban Planning, 126*, 84–93.

Kulp, S. A., & Strauss, B. H. (2019). New elevation data triple estimates of global vulnerability to sea-level rise and coastal flooding. *Nature Communications, 10*(1), 4844.

Kowalewski, M. (2021). Images and spaces of port cities in transition. *Space and Culture, 24*(1), 53–65.

Mahoney, J. (2000). Path dependence in historical sociology. *Theory and Society, 29*(4), 507–548.

Nicholls, R. J. (2011). Planning for the impacts of sea level rise. *Oceanography, 24*(2), 144–157.

Nursey-Bray, M., Blackwell, B., Brooks, B., Campbell, M. L., Goldsworthy, L., Pateman, H., & Hewitt, C. L. (2013). Vulnerabilities and adaptation of ports to climate change. *Journal of Environmental Planning and Management, 56*(7), 1021–1045.

O'Riordan, T., Nicholson-Cole, S. A., & Milligan, J. (2008). Designing sustainable coastal futures. *Twenty-First Century Society, 3*(2), 145–157.

Osbaldiston, N. (2018). *Towards a sociology of the coast*. Palgrave.

Palmer, S. (2020). History of the ports. *International Journal of Maritime History, 32*(2), 426–433.

Pham, T. D., Dwyer, L., Su, J.-J., & Ngo, T. (2021). COVID-19 impacts of inbound tourism on Australian economy. *Annals of Tourism Research, 88*, 103–179.

Ports Australia. (2022). *Trade Statistics*. Retrieved from https://www.portsaustralia.com.au/resources/trade-statistics.

Ports North. (2022). *Trade Statistics*. Retrieved from https://www.portsnorth.com.au/trade-property-development/trade-statistics/.

Rahmstorf, S. (2010). A new view on sea level rise. *Nature Climate Change, 1*(1004), 44–45.

Shaw, C. (2010). The dangerous limits of dangerous limits: Climate change and the precautionary principle. *The Sociological Review, 57*(2), 103–123.

Shove, E. (2010). Social theory and climate change. *Theory, Culture & Society, 27*(2–3), 277–288.

Simmel, G. ([1903]1997). The metropolis and mental life. In D. Frisby & M. Featherstone (Eds.), *Simmel on Culture: Selected Writings* (pp. 174–186). Sage.

Tourism Research Australia. (2021). *State of the industry*. Retrieved from https://www.tra.gov.au/economic-analysis/state-of-the-industry/state-of-the-industry.

Urry, J. (2016). *What is the future?* John Wiley & Sons.

Valente, S., & Veloso-Gomes, F. (2020). Coastal climate adaptation in port-cities: adaptation deficits, barriers, and challenges ahead. *Journal of Environmental Planning and Management, 63*(3), 389–414.

Weber, M. (1915 [1970]). *From Max Weber: Essays in sociology*. Routledge and Kegan Paul.

Weber, M. (1919[2012]). Science as a profession and vocation. In H. H. Bruun & S. Whimster (Eds.), *Max Weber: Collected Methodological Writings* (pp. 335–354). Routledge.

Wynne, B. (2010). Strange weather, again. *Theory, Culture & Society, 27*(2–3), 289–305.